

Review

Revealing the Complexity of Polysaccharides: Advances in NMR Spectroscopy for Structural Elucidation and Functional Characterization

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Abstract: Polysaccharides are important biomolecules, which play a key role in biological, medical, and industrial processes due to their diverse structures and important functions. This paper looks into the significant progress made in the structural and functional analysis of polysaccharides by nuclear magnetic resonance (NMR) spectroscopy. NMR spectroscopy, including solution-state and solid-state technology, has revolutionized this field through detailed molecular insights into the structure, conformation, and dynamics of polysaccharides at the molecular level. There have been some important historical breakthroughs in 1D and 2D NMR, which have led to modern methods like multidimensional NMR and nuclear dynamic polarization (DNP). These modern methods offer a high level of resolution and sensitivity and have made it easier to come up with innovative applications. The applications range from the structural elucidation of microbial and plant structures of polysaccharides to improving food texture, developing therapies, and creating sustainable materials. Despite challenges such as signal overlap and limited sensitivity, emerging solutions are making significant progress. Computational modeling, isotope labeling, and integrated methods that combine complementary technologies are driving the boundaries of polysaccharide research. This review demonstrates the transformative role of NMR in revealing the complexity of polysaccharides and its potential to promote future discoveries and new ideas in the dynamic field.



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1. Introduction

Polysaccharides are basic biological molecules that have attracted scientific attention for decades. Their structural diversity and functional significance in biological, medical, and industrial applications drive ongoing research [1–3]. Historically, early studies on their structures relied on chemical degradation and chromatographic techniques. Scientists used these techniques to figure out their monosaccharide composition and glycosidic bonds [4]. With the progress of analytical technology, nuclear magnetic resonance (NMR) spectroscopy has changed the field by providing direct and non-destructive ways to understand the complex structures of polysaccharides [5,6]. In the 1980s, the breakthrough of 1D NMR spectrum laid the foundation for detailed research on polysaccharides [7]. Subsequently, in the 1990s, two-dimensional NMR technology was developed, such as COSY, HSQC and TOCSY, enabling researchers to analyze overlapping signals and identify glycosidic bonds [6]. For example, Yao et al. summarized different NMR methods used for conformational analysis in a systematic way [8]. However, their review did not include new

technologies like nuclear dynamic polarization (DNP). Similarly, Rodríguez Sánchez et al. concentrated on the structural analysis of sulfate polysaccharides in seaweed [9]. However, they did not deal with the application of NMR in the analysis of complex matrices such as food systems. The aim of this review is to fill these gaps by focusing on three key aspects. First, it assesses the latest progress in NMR technology including DNP, multidimensional NMR techniques, and integrated computational modeling to improve structural analysis. Second, it points out the expanding role of NMR in different fields. For example, in food science, it can help make the texture of food better, and in microbiology, it is useful for understanding how biofilms work. Finally, we investigated the complementary use of mass spectrometry and molecular dynamics simulations, which improved the accuracy and depth of polysaccharide structure studies. The review provides an interdisciplinary perspective and comprehensively summarizes the development of NMR in polysaccharide science by combining academic research and industrial applications.

The development of cross-polarization magic-angle spinning (CP/MAS) and DNP techniques has made the sensitivity of ssNMR better, allowing scientists to distinguish the difference between crystalline and amorphous regions in polysaccharide samples [10]. In addition, using advanced isotope labeling strategies like ^{13}C enrichment has made it easier to perform a detailed structural analysis of microbial polysaccharides and how they interact with host cells [11,12]. These advances also expand the scope of polysaccharide research to areas like biofilm formation, food texture improvement, and the development of therapies based on polysaccharides [13–15].

The goal of the review is to put these advances together, focusing on how magnetic resonance spectroscopy can be used to figure out the structure, conformations, and dynamics of polysaccharides. The article provides an overview of the past development and basic methods of polysaccharide research using NMR. It thoroughly studies the solution-state and solid-state NMR methodologies, showing examples of using these techniques to figure out the complex structures and functional interactions of polysaccharides. The review thoroughly examines the application of NMR in biomedicine, food, and environmental science to highlight its revolutionary impact. It also explores challenges such as signal overlap and limited sensitivity, which drive advancements in NMR spectroscopy for polysaccharide research. By analyzing the key findings, we aim to emphasize the contribution of NMR to polysaccharide research and explore future directions in the developing field.

2. Evolution of NMR Methodologies

As NMR methodologies evolved from simple 1D experiments to advanced multidimensional approaches, they became increasingly optimized for specific polysaccharide states and research requirements. Table 1 provides a unified classification of these techniques, organized by their applicability to solution- or solid-state samples, while highlighting their unique capabilities for polysaccharide characterization.

Table 1. NMR techniques in solid-state and solution-based samples for polysaccharides.

Sample State	Techniques	Key Features	Optimal Applications
Solution-state	1D $^1\text{H}/^{13}\text{C}$	Rapid chemical shift assignment; simple operation	Purity assessment, monosaccharide fingerprinting (e.g., herbal polysaccharides)
	2D COSY/TOCSY	^1H - ^1H coupling networks; intra-ring connectivity	Homopolysaccharide validation (e.g., starch, cellulose)
	2D HSQC/HMBC	^1H - ^{13}C correlations; glycosidic linkage mapping	Heteropolysaccharide structure (e.g., microbial EPS)
	DOSY	Diffusion-based component separation; Hydrodynamic radius	interactions in mixtures (e.g., pectin–protein complexes)

Table 1. Cont.

Sample State	Techniques	Key Features	Optimal Applications
Solid-state	CP/MAS	Resolves rigid molecular structures; crystallinity analysis	Plant cell wall polysaccharides (cellulose–lignin interactions)
	DNP-NMR	10–100× sensitivity enhancement; low-abundance components	Biofilm matrix, fungal cell walls
	HETCOR	Heteronuclear correlations; hydration studies	Food gel hydration mechanisms

2.1. Solution-State NMR for Polysaccharides

Solution-state NMR techniques have proven to be important in polysaccharide research due to their ability to provide detailed information about the molecular composition, structure, and dynamics in the solution environment. These methods allow researchers to examine complex polysaccharides without requiring crystallization, making them a great choice for studying natural heterogeneous systems. Researchers can identify glycosidic bonds, branching patterns, and correlations between atoms by employing advanced 2D NMR experiments, including COSY, HSQC, TOCSY, and NOESY [16–18].

Recently, solution-state NMR has been applied in many fields, including the characterization of microbial extracellular polysaccharides. It is also helpful for identifying bioactive polysaccharides in metabolomics. Moreover, it can be used to evaluate polysaccharide interactions in traditional drugs and food matrices [19–21]. These advances show that the latest NMR is very important in bringing together basic research and practical uses in polysaccharide science.

2.1.1. Structural Elucidation

The structure of a polysaccharide can be elucidated using NMR spectroscopy, revealing its chemical composition and molecular arrangement. For example, Marvelys et al. used COSY and HSQC to analyze the structure of *Sterculia apetala* gum polysaccharides and reveal the complex branching and functional groups of these polysaccharides [16]. Recognizing these branching patterns is essential, as they can have an impact on the rheological properties of polysaccharides, which are crucial for the applications in both the food and medical industries. These studies highlighted the power of 2D NMR in uncovering structural details that are hard to figure out with traditional methods.

Similarly, Cai et al. used ^1H and ^{13}C NMR to characterize the polysaccharides of *Streptococcus pneumoniae* serotype 6C. They identified the immunogenic epitopes of these polysaccharides and found their potential uses in vaccine development [17]. The novel aspect of their work is that they combined structural research with biomedical applications, demonstrating how NMR can connect basic research and translational science. By combining structural analysis with translational research, they pointed out that NMR has the potential to be an innovative discovery tool in glycobiology.

Advanced technologies like HMBCs and NOESY have further improved structural elucidation. In a study about bacterial polysaccharides, Naumenko et al. combined 1D and 2D NMR with methylation analysis to decode the O-antigen structure of *Escherichia albertii*, highlighting the versatility of NMR in microbial polysaccharide studies [18]. Integrating different technologies reveals a novel way to deal with the challenges of complex and overlapping signals. The methodology developed in this study represents a contemporary approach for resolving signal overlap challenges in structurally diverse microbial polysaccharides, like *Escherichia albertii* O-antigens. These results provide a framework that can be copied to solve similar problems in glycobiology.

The structural elucidation of polysaccharides, especially those derived from medicinal plants like *Radix Astragali* (RAPS-3), as illustrated in Figure 1, presents significant chal-

lenges due to their complex and heterogeneous structures [21]. For example, in RAPS-3, ^1H NMR spectroscopy alone revealed the presence of a backbone with α -(1 $\rightarrow$ 4)-linked D-glucan. However, the exact nature of the glycosidic linkages and the configuration of side chains could not be fully confirmed without further 2D NMR analysis. The application of 2D NMR experiments, specifically COSY, HSQC, and HMBC, was crucial to elucidating the detailed structure of these polysaccharides. In the HSQC spectrum of RAPS-3, the anomeric proton at δ_{H} 5.42 (H-1) showed a direct correlation with the carbon signal at δ_{C} 102.5, confirming the α -(1 $\rightarrow$ 4) linkage between glucose units in the backbone of the polysaccharide. HSQC also helped confirm the identity of side chain components like arabinose, as cross-peaks between the anomeric protons of arabinose at δ_{H} 5.20, 5.17, and 5.11 and their corresponding carbons at δ_{C} 109.8, 109.7, and 110.3 were clearly observed. The HMBC spectrum provided long-range correlations between protons and carbons that are separated by two or more bonds, helping to definitively establish glycosidic linkages. For instance, a significant cross-peak between the anomeric proton at δ_{H} 5.42 and the C-4 carbon at δ_{C} 79.6 in the HMBC spectrum confirmed the α -(1 $\rightarrow$ 4) linkage in the glucan backbone. Additionally, the cross-peak between H-4 (δ_{H} 3.68) and the anomeric carbon (δ_{C} 102.5) further substantiated the linkages within the glucan backbone. These 2D NMR experiments allowed for a detailed and unambiguous structural elucidation of RAPS-3, revealing a backbone of α -(1 $\rightarrow$ 4)-D-glucan with side chains of arabinose, as well as confirming the presence of other minor polysaccharide units. The ability to distinguish between different anomeric protons (e.g., from arabinose and glucose) and identify their respective linkages to the main polysaccharide chain is a direct advantage of 2D NMR, making it indispensable for structural analysis of complex polysaccharides like those from *Radix Astragali*. Hence, 2D NMR techniques such as COSY, HSQC, and HMBC are critical in overcoming the challenges of structural elucidation of polysaccharides. They provide the necessary resolution to distinguish between different sugar units, determine the exact nature of glycosidic linkages, and confirm the overall architecture of polysaccharide chains, which is essential for understanding their biological activities and applications in medicinal studies.

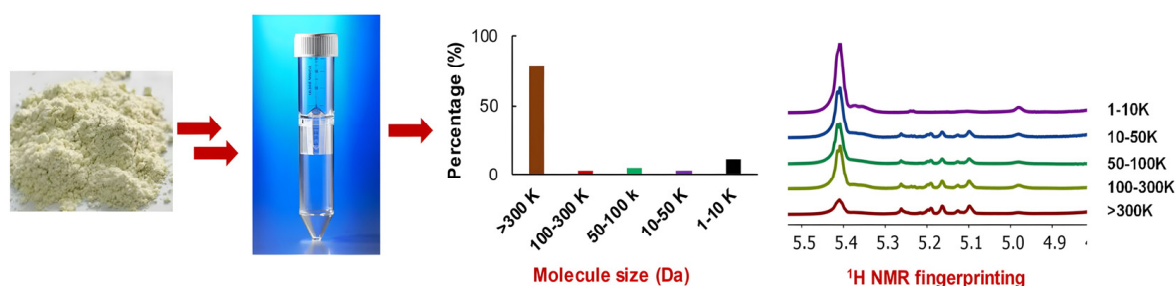


Figure 1. Schematic illustration of rapid preparation and proton NMR fingerprinting of polysaccharides from *Radix Astragali*. Copyright (2025) Elsevier.

2.1.2. Conformational Dynamics

Besides static structural information, NMR provides dynamic insights into polysaccharide flexibility and conformational behavior. A main advantage of NMR lies in its ability to probe molecular motions across multiple timescales, providing a comprehensive picture of conformational changes and interactions. For example, Poveda et al. used ROESY and T_1/T_2 relaxation measurements to study the conformational dynamics of extracellular polysaccharides in slow-growing rhizomes from *Bradyrhizobium*. They found out how molecular flexibility affects biological functions [22]. The paper emphasizes how the dynamic features of polysaccharides influence their ability to interact with other biomolecules like proteins and receptors. It shows the functional role of polysaccharides in biological

systems. NMR-based technologies, like NOESY and relaxation time experiments, can also help us figure out transient conformations. These conformations play a part in biological activity. For instance, Vlachou et al. demonstrated the synergy between NMR and MD by using MD simulations to validate NOESY results, offering dynamic insights into transient hydrogen bonds of scleroglucans. These simulations helped confirm the formation and stability of the hydrogen bonds observed in the NMR experiments. The MD simulations allowed the researchers to explore the flexibility and transient nature of these hydrogen bonds, which were difficult to capture in static NMR data alone. By simulating the molecular dynamics of glucan in different environments, the MD results complemented the NMR findings, giving a deeper understanding of the behavior of hard glucans under physiological conditions [23].

Martin-Pastor et al. compared the conformational dynamics of polysaccharides and their repeating oligosaccharide units. They showed how molecular size affects elasticity and interactions [24]. This comparative method highlights that molecular structure is important in determining dynamic behavior. These studies give us basic insights into the conformation of polysaccharides. They also allow us to rationally design materials with specific mechanical or biological properties.

Recent progress in computational modelling, when combined with NMR data, has further enhanced our ability to analyze conformational dynamics. For instance, researchers used a combination of molecular dynamics simulation and dipole coupling (RDC) measurement. They did this to improve the conformation model of polysaccharides. It also increased the accuracy of predicting the functional role of polysaccharides [6,15].

This integration is a very promising way to study polysaccharides in more and more complex biological systems. Clement et al. used a combination of molecular modeling and NMR spectroscopy to investigate the conformations of pentose fragments derived from the O-specific polysaccharide of *Shigella flexneri* 5a. The NMR study, utilizing ^1H and ^{13}C chemical shifts, inter-residue distances, and heteronuclear coupling constants, revealed that one of the pentasaccharides closely mimics the conformation of the native polysaccharide in solution. Molecular modeling, carried out using the CICADA algorithm and MM3 force field, allowed for a detailed conformational search, which was validated by the NMR data. These findings demonstrated functional and structural correlations, particularly showing how specific structural features of the pentasaccharides relate to their biological recognition by monoclonal antibodies targeting *Shigella flexneri* 5a [25].

Coxon et al. utilized MD simulations to refine the conformation of an octasaccharide molecule by integrating RDC data. The RDC data provided long-range orientational restraints, which were then incorporated into the MD simulations. These simulations were essential in optimizing the sugar molecule's three-dimensional structure, allowing for more accurate conformational predictions than those based on RDC data alone. The integration of MD with NMR data helped resolve ambiguities in the sugar's conformation, enhancing the understanding of its structural dynamics [26]. MD simulations complemented NMR by providing dynamic perspectives, helping to validate and optimize structural and dynamic information. This integration allowed for a more comprehensive and accurate molecular model by combining the strengths of both experimental and computational methods.

2.1.3. Applications in Complex Mixtures

NMR in the solution state is very good for analyzing polysaccharides in complex mixtures, like those in food matrices and biological samples. The reason is that it can detect and distinguish molecular components in their natural state without requiring extensive sample preparation. For instance, Merks et al. measured mixtures of food polysaccharides using ^1H NMR, and these measurements can be used for quality control

and process optimization [19]. Their research showed that NMR can analyze multiple polysaccharide components at the same time. This provides a reliable way to characterize heterogeneous systems.

In the pharmaceutical field, de Carvalho et al. used diffusion-ordered NMR spectroscopy (DOSY) along with methylation and advanced 1D/2D magnetic resonance techniques (like HSQC, HMBC, TOCSY) to analyze the structural characteristics of β -glucan from yeast and mushrooms, as shown in Figure 2 [27]. The main interesting thing about this paper is that it uses DOSY to tell different types of polysaccharides in complex mixtures apart by analyzing their diffusion behavior. This method provides useful information about the patterns and structures of macromolecular chains. These patterns and structures are hard to find with traditional techniques. By using this method, DOSY demonstrates its role as a powerful complementary tool for polysaccharide structural analysis. It also has the potential to show details at the molecular level. These details are very important for us to understand the biological functions of polysaccharides.

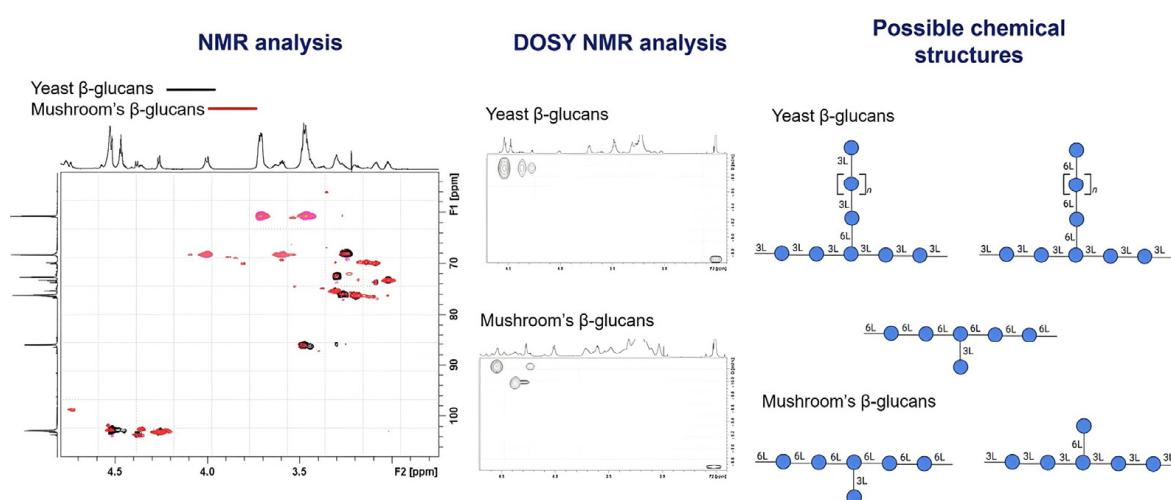


Figure 2. The structural characteristics of polysaccharides from yeast and mushrooms using diffusion-ordered NMR spectroscopy (DOSY). Copyright (2025) Elsevier.

NMR-based metabolomics is also helpful for the study of polysaccharides in biological matrices. Li et al. studied the polysaccharides from *Polygonatum sibiricum* in traditional medicine using ^1H NMR. They identified the bioactive components and connected them to observed health benefits [20]. This method combines structural analysis and functional studies. It helps us understand how specific polysaccharide components promote biological activity. Chen et al. used ^1H NMR-based metabolomics to investigate the antidiabetic activity of *Enteromorpha prolifera* polysaccharides and proved that they have therapeutic potential [28]. Likewise, De Souza et al. developed a general NMR method to analyze the polysaccharide composition, providing a reliable means for structural characterization [29]. In addition, Ye et al. analyzed the polysaccharide part of glycopeptides and found the uses of these drug structures [30].

Wang et al. utilized graded acid hydrolysis and NMR to elucidate the structure of uronic-acid-containing polysaccharides from abalone, which demonstrated how sequential hydrolysis can aid structural studies [31]. This research illustrates that NMR-based metabolomics can be a very powerful tool for linking polysaccharide structures to their biological and pharmaceutical functions. This integrative approach not only helps us understand polysaccharides better but also creates new ways to use them in therapeutic innovation and drug design.

Cordeiro et al. used NMR technology to monitor polysaccharides during wine fermentation. Their research showed the changes in polysaccharide components during fermentation. The study presents a novel approach to improve wine quality by managing polysaccharides [32]. Falourd et al. studied how polysaccharides interact with water by dynamic vapor adsorption and NMR to understand hydration kinetics at the molecular level [33]. Wu et al. characterized polysaccharide interactions in bilateral mixtures with low-field NMR. They also pointed out the application of this research in food and polymer research [34].

These examples illustrate NMR's versatility in analyzing polysaccharides in complex and dynamic environments, from food to biological and industrial systems. By performing structural studies and developing practical applications, NMR continues to expand our understanding of polysaccharide behavior in real-world mixtures.

2.2. Solid-State NMR for Polysaccharides

Solid-state NMR (ssNMR) has become an important method for studying polysaccharides due to it providing detailed information about how rigid and semi-rigid molecules are arranged. Scientists can use advanced technologies like CP/MAS and DNP with ssNMR. These technologies allow them to detect the molecular structure, crystallinity, and dynamic behavior of polysaccharides in different systems. These methods have been shown to work very well when studying plant cell walls, biofilms, and hydrogels, where polysaccharides have different levels of crystallinity and amorphous regions [2].

The broader application of ssNMR in polysaccharide research has helped us understand the molecular system and functional properties better. For example, ssNMR is very crucial for telling the difference between crystalline and amorphous regions, allowing us to analyze the mechanical and functional properties in detail. In addition, when we combine ssNMR with other technologies, it can be expanded in more areas, offering a comprehensive way to deal with the problems of polysaccharide systems.

2.2.1. Crystalline and Non-Crystalline Polysaccharides

Solid-state NMR is a crucial technique for distinguishing the difference between the crystalline and amorphous regions of polysaccharides, offering detailed information about their molecular structure. This distinction is particularly important in plant cell walls, where the mechanical properties depend on the balance between crystalline and amorphous regions. For example, Zujovic et al. used CP/MAS NMR to compare the polysaccharide structures in the thick-walled corneum tissue and the thinner-walled celery cell walls. Their study revealed structural differences that were linked to the mechanical functions of these tissues [2]. Bonanomi et al. used ^{13}C CP/MAS NMR spectroscopy to study the molecular composition of organic matter in many biological kingdoms. These kingdoms, including bacteria, fungi, algae, and higher plants, are illustrated in Figure 3 [35]. The key point is that the largest publicly available library containing 108 unique NMR ^{13}C spectra has been created. These spectra can clearly indicate the difference between carbon types. They can also show how carbon is distributed among different species and functional groups. This method can show the different chemical properties of organisms from different kingdoms. It also points out the functional differences in plant tissues like leaves, roots, and wood. The analysis yields critical insights into the carbon cycle and decomposition processes in the ecosystem.

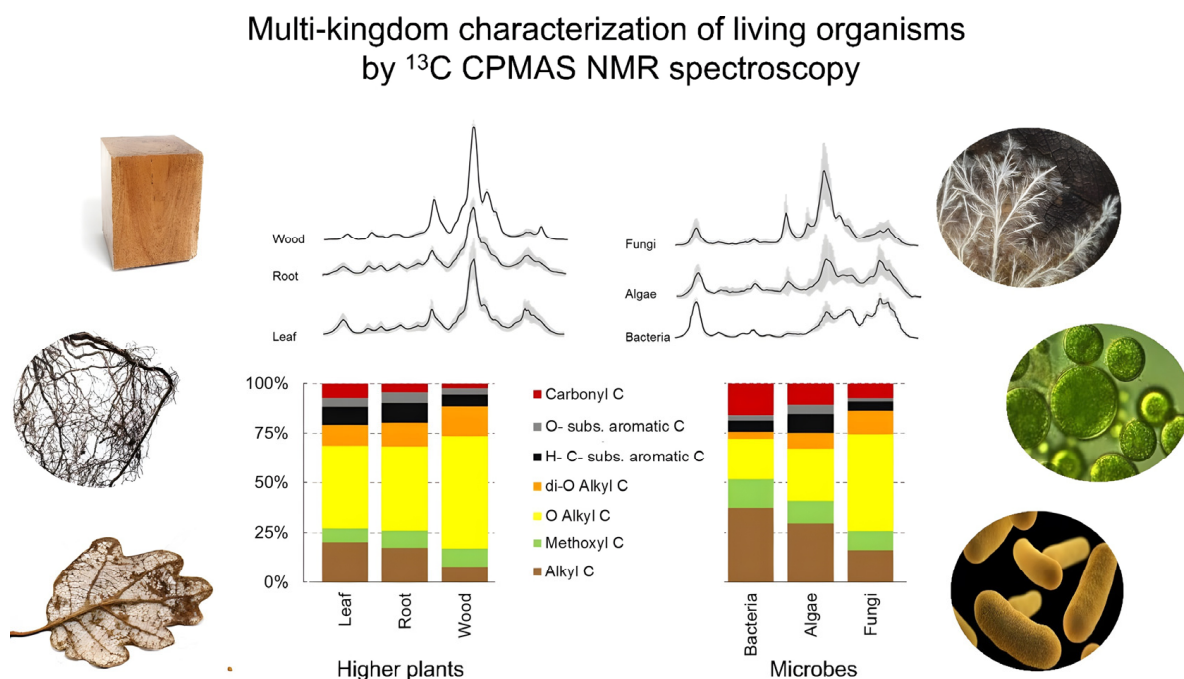


Figure 3. Solid-state ^{13}C CP/MAS NMR spectroscopy to characterize the molecular composition of organic matter across multiple biological kingdoms, including bacteria, fungi, algae, and higher plants. Copyright (2024) Elsevier.

Delcourt et al. used magic-angle spinning NMR to study the spectra of polysaccharides in whole cells, improving the clarity of the cell wall's structure and understanding of its functions [36]. These examples show that solid-state NMR has made an important contribution to elucidating the complex structural organization of polysaccharides in different biological systems. By combining structural details with functional insights, ssNMR continues to enhance our understanding of the relationship between molecular architecture and biological properties.

The innovation of DNP-enhanced NMR has greatly improved the sensitivity of ssNMR technology. This improvement allows researchers to study the structure of low-abundance polysaccharides. Xue et al. used this method to study wood polysaccharides during termite fermentation. Through this study, they obtained important insights into the enzymatic degradation mechanisms related to biofuel production [10].

In food science, Ng et al. used ssNMR technology to study how the crystallinity and amorphous regions of apple polysaccharides work. They showed how these differences in structure impact the texture and ripening processes of apples [37]. Davies and his group also found that the crystal cellulose structure reorganized after the extraction of polysaccharides. These findings demonstrate an idea of how processing affects the composition of plant cell walls [38]. Cheng et al. utilized ssNMR technology to investigate the molecular structure of fungal cell walls, revealing the complex structures of chitin and chitosan. They also discovered the interactions between rigid and mobile domains. The technique provides insights into the organization of cell walls at the atomic level and helps to solve the problems of understanding the complex structure and dynamics of cell walls [39]. Adam et al. completed structural elucidation of capsular polysaccharides from *Klebsiella* I-714 with advanced NMR technology in industrial applications [40]. These studies together show the versatility of ssNMR in unraveling the complex relationships between structure and function in polysaccharides, enabling applications across agriculture, food science, and renewable energy research.

2.2.2. Interactions with Water and Small Molecules

Water interactions play a critical role in polysaccharide functionality, particularly in hydration, gelation, and swelling. NMR spectroscopy, especially ssNMR, has proven to be a valuable tool for studying these interactions. For example, Falourd et al. combined time-domain NMR with dynamic vapor adsorption to study how polysaccharides and water interact, revealing the kinetics of hydration at the molecular level [41]. Bellich et al. analyzed biofilm polysaccharides and their interactions with quorum-sensing molecules, providing an understanding of biofilm dynamics and bacterial communication [42].

Agles and Bourg employed NMR spectroscopy alongside MD simulations to investigate the interactions between polysaccharides, such as alginate, and water. The NMR spectra, specifically ^1H and ^{13}C NMR, were instrumental in identifying how water molecules interact with polysaccharide matrices at the molecular level [43]. These NMR analyses allowed the researchers to map the distribution of water within the polysaccharide network and distinguish between water molecules in close proximity to the polysaccharide (bound water) and those farther away (free water). The shift in chemical shifts and relaxation times observed in the NMR data provided insight into the dynamic behavior of water, particularly how the polysaccharide matrix influences the water's structure and mobility. By integrating these NMR results with MD simulations, Agles and Bourg were able to correlate the structural features of polysaccharides with their effect on water dynamics, offering a comprehensive picture of how these biopolymers interact with water at both the molecular and macroscopic levels. This combined approach of NMR and MD simulations not only sheds light on the detailed structural interactions but also enhances the understanding of polysaccharides' roles in water-related biological processes, providing valuable insights into their potential in biomedical applications such as drug delivery and tissue engineering. Similarly, Baranowska et al. explored the interactions between potato starch and colloids with low-field NMR, showing how to optimize the texture properties of food science [44]. The paper also emphasizes that water binding is important in determining the rheological properties of polysaccharides, which plays a crucial role in food and pharmaceutical preparations.

Based on these findings, Juris et al. used ssNMR to study the interactions between wheat starch and *Mesona chinensis* polysaccharides. Their research found out specific ways of hydration that have an impact on gel network formation. It also provided an understanding of how polysaccharides influence the structural properties of complex food matrices [45]. In another study, Savary et al. examined the diffusion of small molecules within composite polysaccharide gels using rheological data combined with ssNMR. This integrated approach allowed the researchers to study the relationship between the gel composition and molecular fluidity. By doing so, they can figure out the characteristics of the microstructure of the polysaccharide aqueous gel [46]. These studies emphasize the versatility of ssNMR in probing polysaccharide–water interactions and its growing role in applications ranging from food science to materials engineering. By combining ssNMR with complementary methods, researchers can achieve a more holistic understanding of hydration and gelation processes in polysaccharides.

2.3. Multidimensional and Advanced NMR Techniques

Multidimensional and advanced NMR techniques have significantly expanded the scope of polysaccharide research by overcoming the limitations of one-dimensional methods and enhancing spectral resolution. These methods are very important for studying complex molecular structures. They can also link structural modifications with functional results. Moreover, they can show the dynamic behavior in heterogeneous systems. For example, 2D and 3D experiments like COSY, HSQC, TOCSY, and NOESY, can

distinguish overlapping signals. In addition, advanced methods such as ordered diffusion spectroscopy (DOSY) are available for characterizing molecular interactions and diffusion dynamics [6,47].

Beyond structural elucidation, multidimensional NMR is also useful for studying how molecules in polysaccharides interact. For example, it can be used to study the gel, hydration, or stable interactions in food matrices. Techniques like DNP-enhanced NMR have also greatly improved the sensitivity. This improvement allows us to study the low-abundance components in complex systems [5,8]. These advances have made multidimensional magnetic resonance a key part of polysaccharide research. It combines research at the molecular level with practical uses in food, biomedical, and environmental sciences.

2.3.1. Multidimensional NMR

Multivariate NMR experiments, like 2D HSQC-TOCSY and 3D NOESY-HSQC, offer extremely good resolution for complex polysaccharides. Scientists can use these techniques to resolve overlapping signals and determine long-range correlations. To systematically compare the advantages and limitations of key multidimensional NMR techniques (HSQC vs. HMBC vs. NOESY) for polysaccharide analysis, we summarize their characteristics in Table 2. These findings are pivotal for understanding the structures of complex polysaccharides. For example, Kou et al. used 2D NMR to characterize the stability of a tamarind seed polysaccharide balsam that was changed with octenylsuccinic anhydride, and they found that it could be a useful ingredient in functional foods [13].

Table 2. Comparison of key NMR techniques in polysaccharides (e.g., HSQC vs. HMBC vs. NOESY).

Technique	Advantages	Limitations	Typical Applications
HSQC	^1H - ^{13}C direct correlation; ideal for glycosidic bond identification and monosaccharide composition	Lower sensitivity for low-abundance samples	Primary structure determination (branching patterns, anomeric configurations)
HMBC	Detects ^1H - ^{13}C long-range couplings (2–4 bonds); reveals cross-glycosidic linkages	Weak signal intensity; requires longer acquisition time; challenging for overlapping signals	Sequence determination in complex polysaccharides (e.g., bacterial O-antigens)
NOESY	Provides spatial proximity information (<5 Å); reveals conformation and intermolecular interactions	Sensitive to molecular motion; difficult to quantify	Conformational studies (e.g., food gels, biofilms)

Kulik and Haverkamp studied molecular mobility in starch polysaccharides using two-dimensional ssNMR spectroscopy, revealing dynamic properties critical to food applications [48]. This study demonstrates that multidimensional NMR can uncover subtle changes and link them to functional characteristics. Leeflang et al. used 3D NOESY-HSQC to study polysaccharides and glycoprotein polysaccharides. They managed to improve the spectral resolution and identify unique structural patterns that influence biological activity [47]. Their work shows that multidimensional NMR is important for connecting the structure and function of bioactive polysaccharides. In addition, Jansson et al. used CASPER along with multivariate NMR data to determine the sequence and structure of complex oligosaccharides [6]. Fontana et al. demonstrated that combining computational tools with NMR data can be used to analyze O-antigen polysaccharides, further expanding the practical use of multidimensional NMR in explaining structures [49]. Kupce et al. brought up the problem of resolving overlapping signals. They especially focused on

advanced methods for decoding the nuclear Overhauser effect [50]. This combination of computational analysis and experimental NMR data shows that multidimensional methods have potential in automated structure elucidation.

Kolz et al. used multidimensional relaxation time NMR to study dual mixtures of liquids and polysaccharides. Through this study, they developed some ideas about the molecular fluidity and behavior of the mixture [51]. These examples illustrate that multidimensional NMR technology has become very important in polysaccharide research, particularly for dealing with the complex structures of polysaccharides and studying the relationship between their structure and functions.

As the resolution and sensitivity of multidimensional NMR continuously improves, it is paving the way for new discoveries in polysaccharide science fields like food science, biomedical science, and environmental science. This study is notable for the application-oriented study, demonstrating how structured analytics can offer information for product development.

2.3.2. Advanced ssNMR for Dynamics and Interactions

Advanced ssNMR techniques, including proton spin diffusion, heteronuclear correlation experiments (HETCOR), and advanced dipole coupling analysis, have enhanced detailed studies on molecular mobility and interactions.

These methods are very helpful when it comes to studying the interactions between structure and dynamics in complex polysaccharide systems. Kirui et al. used advanced ssNMR technology to investigate the primary cell wall of *Arabidopsis*. They demonstrated how pectin methyltransferase regulates polysaccharide interactions and dynamics inside the cell wall matrix [52]. Their findings show the role of a dynamic assembly of polysaccharides in determining the mechanical properties of cell walls. This helps us better understand plant biology.

Another application is a combination of ssNMR time-resolved ssNMR with DNP. Fernando et al. studied fungal polysaccharides and plant cell walls using DNP-enhanced ssNMR and found subtle changes in molecular interactions under different environmental conditions [5]. This method provided unprecedented sensitivity, paving the way for the study of low-abundance components. Gautam et al. used DNP-enhanced NMR and achieved a 27-fold boost in signal sensitivity, and this made it possible to analyze cell wall polysaccharides in *Aspergillus nidulans* and *A. fumigatus* [8].

As shown in Figure 4, they introduced stable double free radicals (e.g., AMUPol) into the sample, and then used microwave radiation to irradiate it at a low temperature (around 100 K). By doing this, the authors significantly enhanced signal detection, allowing them to identify the weak interactions between polysaccharides, such as chitin and α -1,3-glucan, in the fungal cell walls. The combination of MAS-DNP, long-range 2D NMR, and hydration kinetics provided an unprecedented view of the fungal cell wall structure at the molecular level. It allows us to detect subtle structural differences and interactions that traditional techniques could not detect before.

In the food science field, Yuris et al. used ssNMR to monitor changes in molecular fluidity when a gel is forming. They also studied the interactions between wheat starch and *Mesona chinensis* polysaccharides [45]. This study offered an understanding of the structural properties of a polysaccharide gel, which is important for food optimization. In general, these examples show the transformative potential of advanced ssNMR techniques in elucidating polysaccharide dynamics and interactions across diverse applications. If combined with other methods like isotope labelling and computational modelling, the field will make more progress in understanding complex polysaccharide systems. These methodological improvements set the basis for dealing with the complexity of polysaccha-

rides. In the following parts, we will outline their revolutionary influence in biomedical, food, and environmental applications.

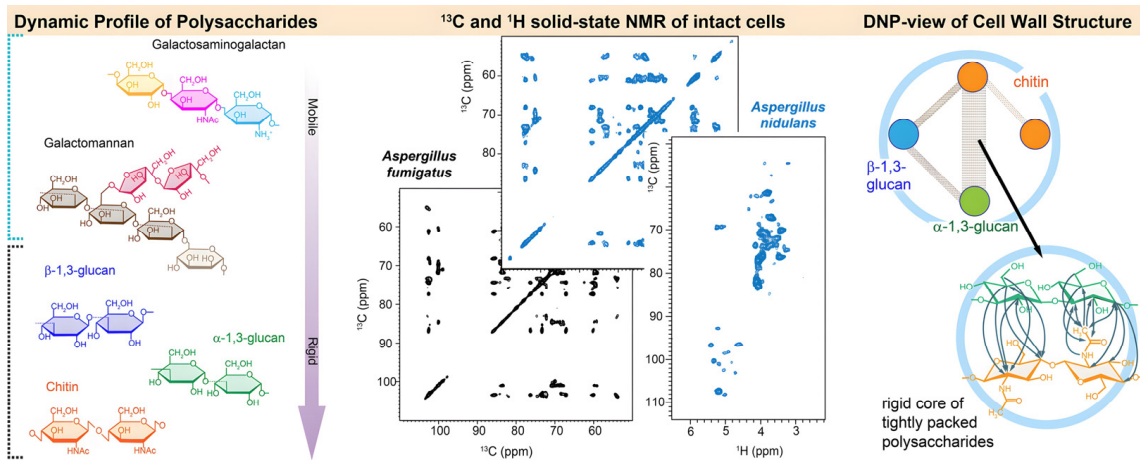


Figure 4. Overview of ^{13}C , ^{15}N , and ^1H -detection solid-state NMR, supplemented by DNP, to compare the structural organization of cell wall polymers and their assembly in the cell walls of *A. fumigatus* and *A. nidulans*. Copyright (2025) Elsevier.

3. Applications of NMR in Polysaccharide Research

The applications of NMR in polysaccharide research span many areas, indicating flexibility and strong functions as an analytical tool. NMR has played an important part in elucidating the structure–function relationships of polysaccharides, contributing to the development of biomedical, food, and environmental sciences.

For instance, it has provided insights into the immunomodulatory and anticoagulant activities of polysaccharides, which promotes therapeutic innovation [15,53]. In the food industry, NMR is vital for studying how polysaccharides interact in complex arrays. This interaction can improve the texture, stability, and sensory properties of food [19,54].

The NMR analysis of marine microorganisms and polysaccharides is beneficial for environmental applications, providing solutions for biofilm management and sustainable technologies [10]. These various applications show that NMR plays a crucial role in turning structural insights into practical benefits in many fields.

To summarize these diverse applications, we have compiled key studies into Table 3. This table outlines the major contributions of NMR in biomedical applications, food science, and environmental research. It highlights key findings, methodologies, and potential impacts in each field.

Table 3. Applications of NMR in polysaccharide research.

Field	Key Findings	Methodology	Potential Impact
Biomedical	Elucidation of immunomodulatory and anticoagulant polysaccharides	Solution-state NMR, HSQC, NOESY	Development of therapeutics for immune modulation and clotting
	Structural characterization of bacterial capsular polysaccharides for vaccines	Multidimensional NMR, isotopic labeling	Improved vaccine design and efficacy
	Antioxidant activity of polysaccharides	^1H NMR, MD	Reduced oxidative stress; therapeutic innovation

Table 3. Cont.

Field	Key Findings	Methodology	Potential Impact
Food Science	Analysis of polysaccharide interactions in food matrices	^1H NMR, DOSY	Optimization of texture, stability, and sensory properties
	Study of gelation and hydration dynamics	Solid-state NMR, relaxation measurements	Enhanced food product formulation and quality control
Environmental	Investigation of microbial exopolysaccharides in biofilms	Solid-state NMR, DNP-enhanced NMR	Development of sustainable bio-film management technologies
	Analysis of lignocellulose degradation in biofuel production	CP/MAS NMR	Advancement in renewable energy technologies

3.1. Biomedical Applications

NMR has been instrumental in studying polysaccharides with biomedical relevance, offering a detailed understanding of their structure–function relationships and guiding therapeutic innovations. Although earlier reviews have given an understanding of the particular uses of NMR in polysaccharide research, our review is different, combining the recent progress in multidimensional NMR, DNP, and computational modeling techniques. This helps to deal with the current challenges and creates new research directions. For example, Mulloy et al. elucidated the structure of anticoagulant sulfate polysaccharides. They also gained a deeper understanding of how they work and how they interact with coagulation factors [15]. This work demonstrates how NMR can provide insights into the relationship between structural features and biological activity. It also lays the foundation for the development of anticoagulant drugs.

Yang et al. described an immunomodulatory polysaccharide that comes from *Candida albicans*, linking its structural motif with the activation of the immune system [53]. This study highlights the role of NMR in identifying the functional domains that help with biological activity, allowing for the reasonable design of immune-boosting therapies. Similarly, a study by Qin et al. demonstrated that NMR can measure the main molecular features of bacterial capsule polysaccharides used in vaccine formulations. Structural data are crucial in ensuring that the vaccine works effectively and is safe [55]. The examples presented highlight the potential of NMR as a valuable tool for understanding the link between polysaccharide structure and its biological functions. NMR has also provided key insights into the relationship between structure and function. NMR facilitates the development of innovative therapeutics and enhances the precision of biomedical applications.

Liu et al. demonstrated that a polysaccharide (AERP) extracted from *Astragalus membranaceus* residue significantly improved cognitive dysfunction in diabetic mice. This effect was mediated through modulation of the gut microbiota and enhancement of short-chain fatty acids (SCFAs), including butyrate, which are known to exert antioxidant and anti-inflammatory effects by reducing reactive oxygen species (ROS) accumulation and improving mitochondrial function. Importantly, NMR spectroscopy played a central role in characterizing AERP. Through ^1H and ^{13}C NMR, the α -1,4-glucan backbone and branched arabinogalactan structures were elucidated, revealing key structural features likely responsible for AERP's bioactivity. These findings have highlighted the critical function of NMR in linking structural motifs to antioxidant mechanisms, thereby facilitating the understanding of structure–function relationships in bioactive polysaccharides.

In cancer research, Hoang et al. used NMR spectroscopy to study brown algae polysaccharides, which have antioxidant and anticancer properties [56]. The ability to understand molecular interactions and dynamic behaviors in these systems has created new methods for the development of cancer treatments based on polysaccharides.

Davidson et al. also employed ROESY and ^{13}C NMR techniques to distinguish the difference between d- and l-rhamnose in polysaccharide motifs. They did this to understand the specific functions of these sugars better [57]. Tajima et al. analyzed the water-soluble polysaccharides made by *Acetobacter xylinum*. They also found novel insights into the pathways of polysaccharide biosynthesis [58]. Rustandi and Xu used NMR to study the reactivity of derivative polysaccharides, advancing vaccine production through molecular characterization [59]. Petersen et al. used advanced NMR technology to elucidate the structure of *Streptococcus pneumoniae* capsule polysaccharides, providing potential applications in disease prevention [60]. Li et al. fully analyzed the polysaccharides of the *Streptococcus pneumoniae* 15F serotype capsule using advanced NMR technology, demonstrating its effect on immunological applications [61].

Jones and Lemercinier gave approval for NMR to analyze the capsule polysaccharides used in vaccines. They linked the structural features to the quality of the drugs [62]. This research shows that NMR plays an important role in making progress in different fields, ranging from cancer treatment to vaccine development and polysaccharide biosynthesis. Through detailed structural and functional analysis, NMR has become an indispensable tool for combining basic research with practical biomedical innovations.

3.2. Food Science and Nutrition

In food and nutrition science, NMR has been used to analyze polysaccharides in complex arrays, study their interactions with other components, and assess their functional properties. Beeren et al. demonstrated that polysaccharides play a role in stabilizing lotions and gels using NMR spectroscopy, which plays a key role in food preparations [63]. Li et al. studied high- or low-methoxide-value pectin using NMR technology to identify its effect on aromatic compounds in apple juice. Their aim was to figure out the interaction between polysaccharides and volatile compounds [54]. In the same way, Yamazaki et al. analyzed sulfate polysaccharides from *Arthrobacter* species with NMR and methylation analysis. They found that these polysaccharides have potential in food and pharmaceutical formulations [64]. These studies demonstrate that NMR has multiple functions, revealing the structure and functional roles of polysaccharides in the food system. NMR provides critical insights into the interactions and properties of polysaccharides. NMR has also significantly contributed to the innovation and optimization of food formulations and functional ingredients.

NMR is pivotal for us to understand how polysaccharides affect the texture and stability of food. For example, Savary et al. used NMR to study molecular diffusion in polysaccharide gel, enabling a deep understanding of how these structures capture small molecules and affect food consistency [46]. Another study by Yuris et al. investigated the gel behavior of wheat starch and *Mesona chinensis* polysaccharide. They revealed how changes in molecular fluidity during gel formation optimize food texture [45]. Kang et al. characterized rubber polysaccharides using multidimensional NMR. This accelerated their application in food colloids [65]. Dourado et al. elucidated the arabinose structure of immunobioactive pectin polysaccharides, demonstrating their functional role in food and health [66].

Wu et al. characterized the effect of acetylation on non-pectic polysaccharides in mustard mucilage using NMR spectroscopy. By doing so, they advanced its application in food science [67]. This research shows that NMR is very important in finding out the molecular mechanisms of polysaccharide function in food science. NMR provides detailed information about gel, texture, and stability. It keeps supporting innovation in food science and the development of high-performance food ingredients.

In fermented foods, polysaccharides are also important for changing the microbial interactions and product stability of fermented foods. Cordeiro et al. applied NMR technology to monitor the composition of polysaccharides during wine fermentation. This revealed the dynamic changes that affect the sensory properties and quality of the final product [32]. The study showed that NMR can be used to improve the production and quality control of fermented products.

In food science, scientists use NMR to analyze polysaccharides in complex arrays. They also use it to study how polysaccharides interact with other components and to evaluate their functional properties. For example, Li and colleagues investigated the structural modifications of rice starch using NMR spectroscopy, elucidating its physicochemical characteristics within microencapsulation matrices for enhanced probiotic delivery efficacy [68].

In general, these studies show that NMR has many functions in food science, making for design polysaccharide formulations with better functionality, stability, and sensory properties. By offering detailed information about structure and dynamics, NMR is still an essential tool to promote in food science and nutritional research.

3.3. Environmental and Industrial Applications

The environmental applications of NMR in polysaccharide research mainly focus on understanding microbial extracellular polysaccharides, soil stability, and biofilm dynamics. Kokoulin et al. explored O-polysaccharides from marine bacteria [12]. They found that these polysaccharides have the potential to develop biotechnological solutions for environmental challenges. For example, they can be used in bioremediation and biofilm management. Gamian et al. used advanced NMR technology to figure out the structure of bacterial lipopolysaccharides. This helps us understand the composition of biofilms [69]. In the same way, Spevacek et al. used ssNMR to study the molecular order of polysaccharides in biofilms. They obtained a deeper understanding of the structural role of these polysaccharides in microbial ecosystems [70].

In the area of renewable energy, Xue et al. used ssNMR to study how termites enzymatically break down wood polysaccharides. This research provided important insights into the lignocellulose degradation pathways that are involved in biofuel production [10]. The study shows that NMR can connect the structure of polysaccharides with enzyme function, driving progress in sustainable energy technologies. The industrial use of NMR has also been expanded to include the study of hydrogels and composites. These findings are very important for improving hydrogels in drug delivery, agriculture, and tissue engineering.

In agricultural science, Sheng and Cherniak determined the structure of *Clostridium perfringens* capsule polysaccharides. They elucidated the role of NMR in studies of bacterial polysaccharides [71]. Junker et al. used pulsed field gradient NMR to study covalent polysaccharide gel. They pointed out its potential in structural engineering [72]. Li et al. used solid-state NMR to investigate how saliva proteins and polysaccharides interact in calcified structures. This has helped us learn more about dental materials [73]. These examples illustrate that NMR can be used in many ways to solve difficult problems in the environment and industry. When we combine understanding at the molecular level with practical applications, NMR has become an essential tool to promote sustainable technologies and industrial innovation. These examples show that NMR is playing an increasingly important role in dealing with the complex challenges related to the use of industrial polysaccharides.

4. Challenges and Future Directions

Even though NMR spectroscopy has its advantages, there are still some challenges when applied to polysaccharide analysis. Table 4 provides a summary of the challenges

and future directions for polysaccharide research based on NMR. It focuses on the current limitations, emerging solutions, and possible future directions in the field. These tables are meant to provide a concise overview of the key points discussed in the review and make the commentaries easier to read and more useful.

Table 4. Challenges and future directions in NMR-based polysaccharide research.

Challenge	Current Limitations	Emerging Solutions	Future Directions
Signal Overlap	Overlapping signals in complex mixtures	Multidimensional NMR, computational modeling, isotopic labeling	Development of advanced spectral deconvolution algorithms
Sensitivity Limitations	Low sensitivity for dilute samples or small amounts of material	DNP-enhanced NMR, isotopic labeling (^{13}C , ^{15}N)	Integration of DNP with ultra-high-field NMR
Dynamic Processes	Difficulty in studying dynamic polysaccharide–protein interactions in real time	Time-resolved NMR, in situ solid-state NMR	Development of portable NMR systems for real-time monitoring
Sample Heterogeneity	Heterogeneous samples (e.g., plant cell walls) complicate structural analysis	Integration with complementary techniques (e.g., mass spectrometry, X-ray)	Holistic approaches combining NMR with other methods
Accessibility of NMR Instruments	High cost and limited access	Cost-effective NMR systems, open-access NMR facilities	Democratization of NMR technology

One significant limitation is the overlapping of signals in complex polysaccharide mixtures, which can hide structural details, especially in heterogeneous samples. For example, cellulose, hemicellulose, and pectin have overlapping signals. Therefore, analyzing plant cell wall polysaccharides usually requires a large amount of spectral deconvolution [2,52]. To address this issue, advancements in multidimensional NMR techniques, such as ultra-high-field NMR and selective isotope enrichment, are being explored [9]. For example, people have started to use computational tools like machine learning to automate spectral analysis. This has begun to make manual interpretation less complex [6]. Li et al. used nuclear statistical magnetic resonance to study the component diversity of pectin polysaccharides, providing a new framework for the analysis of complex mixtures [74].

Another challenge is the limited sensitivity of NMR when dealing with dilute samples or small amounts of material, as commonly encountered in biomedical applications. However, DNP-enhanced NMR has proven effective in boosting sensitivity, making it possible to investigate low-abundance polysaccharides in biological systems [5,8]. Similarly, scientists have used the labelling of ^{13}C or ^{15}N isotopes to improve the signal resolution and sensitivity in polysaccharide studies, especially in microbial and marine polysaccharides [11,12,75]. In addition, the improved ssNMR methods and other DNP techniques allow researchers to study polysaccharide interactions at the sub-molecular level in complex environments like biofilms [5,10]. Byeon et al. used DNP-enhanced ssNMR to characterize natural bacterial biofilms. DNP can increase the sensitivity by about 75-fold, enabling us to quickly acquire high-resolution 1D and 2D spectra. With these spectra, we can conduct detailed structural analysis of biofilm components. These components include polysaccharides and proteins. We can perform this analysis without having to use isotope labelling or chemical modification. This technology can overcome the sensitivity limits of traditional ssNMR. It also enables us to detect the extracellular matrix (ECM) components in biofilms efficiently. In this way, it reveals new understandings in biofilm tissue [76].

Integrating complementary technologies is important for getting past the current limitations. For example, mass spectrometry can provide information about molecular weight and sequences. On the other hand, X-ray scattering can show us the structural organization. In the structural studies of bacterial capsule polysaccharides, it has been

shown that combining these methods with NMR spectroscopy works well [17,77]. In the future, we should make efforts to improve this integration. Specifically, we should focus on in situ studies of the dynamic processes and interactions of polysaccharide proteins.

The resulting boundary field involves solid-state NMR studies of polysaccharides in intact biological systems like plant tissues and microbial biofilms. These methods can provide a more comprehensive understanding of the composition and function of polysaccharides in their natural environment more comprehensively [5,10]. Additionally, recent advances in portable NMR technology are expected to be used on-site in areas such as food quality control, environmental monitoring, and industrial process optimization. Even though its resolution is currently limited, this technology shows great potential [27].

Finally, improving the accessibility of high-resolution NMR instrumentation is important to expand its applications in polysaccharide research globally. Innovations in hardware design, including cost-effective ultra-high-field systems, could broaden access to this powerful technology, enabling wider use across academic and industrial field.

5. Conclusions

NMR spectroscopy is still an indispensable tool for polysaccharide research, providing great insights into the complex structure, conformation, dynamics, and interactions of polysaccharides. This review shows how the progress in solution- and solid-state techniques, multidimensional NMR experiments, and other complementary methods have completely changed polysaccharide research.

In the field of biomedical sciences, NMR has elucidated critical structure–function relationships, accelerating the development of anticoagulant and immunomodulatory therapy and allowing for the treatment of cognitive impairment through polysaccharide-derived therapy [14,15,53]. These findings demonstrate the potential of polysaccharides in therapeutic design, supported by precise structural validation using NMR spectroscopy.

In food science, NMR has facilitated the optimization of polysaccharide preparations, ranging from improving the stability of lotions and the properties of gels to enhancing the aroma and sensory properties of food research [32,45,54]. These contributions demonstrate that NMR spectroscopy is crucial in improving the quality and functionality of food research.

Environmental and industrial applications have also benefited significantly from NMR technology. This technology reveals a key enzyme mechanism in lignocellulose degradation. It also improves our understanding of biofilm dynamics. Moreover, it supports the design of hydrogels and composites for renewable and agricultural energy [10,12,27]. These studies provide examples of how NMR can combine molecular insights with real-world applications, driving innovation in sustainability and industrial processes.

As NMR technologies continue advancing, their strategic application requires careful matching of technique capabilities to polysaccharide characteristics. Table 5 distills the findings into a practical reference guide, while highlighting emerging opportunities where further methodological development could overcome current limitations (e.g., in situ analysis of dynamic polysaccharide–protein complexes).

In the future, the integration of NMR with complementary approaches, such as mass spectrometry, computational modeling, and DNP-enhanced spectroscopy, will further expand its capabilities. Emerging technologies, including portable NMR systems and ultra-high-field magnets, have the potential to improve accessibility and sensitivity. With these improvements, we can study polysaccharides in their natural state and in the real-world environment [5,6]. Advancements in NMR technology continue to enhance its sensitivity, resolution, and computational capabilities, and therefore the scope of its applications in polysaccharide research. Novel tools like enhanced NMR and spectral analysis based

on machine learning are expected to solve the existing problems. They will also make it possible to study polysaccharides in their natural environment and complex systems. By bridging molecular level insights with practical applications, NMR is set to remain a fundamental technique in polysaccharide science, driving breakthroughs in biomedical, food, and environmental research while promoting innovation and sustainability in these different fields.

Table 5. Recommended NMR techniques by polysaccharide type.

Polysaccharide Type	Structural Features	Recommended Techniques	Rationale
Linear homopolysaccharides (e.g., cellulose)	Simple repeats, rigid chains	Solution: 1D ¹³ C NMR; solid: CP/MAS	1D NMR confirms composition; CP/MAS distinguishes crystalline domains
Branched heteropoly-saccharides (e.g., pectin)	Complex side chains, acidic groups	HSQC (linkages) + HMBC (sequence) + NOESY (conformation)	Multidimensional approach needed for branching and charge effects
Microbial EPS	High heterogeneity, modified groups (e.g., sulfation)	HSQC/TOCSY + DOSY; solid: DNP-NMR (low-abundance)	Combination addresses structural diversity; DNP enhances sensitivity
Food matrix polysaccharides	Multi-component, dynamic interactions	DOSY + ¹ H NMR metabolomics; Solid: HETCOR	DOSY separates components; HETCOR reveals hydration/protein interactions
Biofilm polysaccharides	Semi-rigid, environment-sensitive	Solution: NOESY; solid: DNP-NMR + ¹ H-detection	DNP solves sensitivity issues; NOESY elucidates biofilm formation

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Abbreviations

NMR	Nuclear magnetic resonance spectroscopy
DNP	Dynamic nuclear polarization
CP/MAS	Cross-polarization magic-angle spinning
RDC	Residual dipolar coupling
DOSY	Diffusion-ordered spectroscopy

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